

Shared evidence | Defence and memory cards

Immune memory: recognise, respond, remember

Name _____ Date _____

Models and teaching summaries, not personal medical guidance.

Barrier card

Skin limits entry. Airway mucus traps material and cilia move it towards the throat.

Recognition card

An antigen triggers a specific response. Some lymphocytes develop into antibody-secreting cells.

Memory card

Some lymphocytes persist as memory cells. Later exposure to the same antigen can trigger a faster, larger response.

Chemical defence card

Lysozyme in tears can damage bacterial cell walls. Stomach acid kills many swallowed pathogens.

Antibody card

Antibodies are proteins with specific binding sites. Binding helps neutralise a target or marks it for other immune action.

Vaccine card

Vaccines expose the immune system to antigens, or instructions for making them, to develop protection and memory without needing the full disease. Protection is not a guarantee.

Elapsed day	First units	Later same antigen units
0	0	0
2	0	8
4	2	20
6	6	32
8	10	30
10	7	24

Constructed teaching data: arbitrary antibody units, not clinical results or a protection threshold.

Supported route | Build the explanation

Word bank: barrier; antigen; antibody protein; lymphocyte cell; memory; same; faster.

S1. Choose skin, mucus/cilia or stomach acid. Explain one action.

S2. Label antigen, antibody and memory lymphocyte as target feature, protein or cell.

S3. At day 4, read both amounts and calculate the difference.

S4. Complete: antigen exposure triggers a specific response and _____. A later response to the _____ antigen can be _____. Explain why.

Standard route | Explain, then use evidence

Use the defence cards and the constructed table.

M1. Explain two different barriers: one physical and one chemical.

M2. Explain the roles of antigen, antibody and memory lymphocyte without calling a protein a cell.

M3. Use day-6 data to compare first and later responses. Include the difference and a memory explanation.

M4. Explain how vaccination can prepare a later response. Why is the graph not a guarantee for one person?

Stretch route | Evaluate the model

Optional deeper reasoning. The chart compares the same antigen. Antigen Y is structurally different from X.

T1. Explain why an antibody selected for X may not bind Y. State a limit of a simple shape model.

T2. Compare the time needed to reach at least 20 units in the first and later data. What can and cannot be concluded?

T3. “Antibody levels fell, so all immune memory vanished.” Evaluate this statement.

T4. Contrast an antibiotic with a vaccine by target/mechanism. Avoid a treatment recommendation.

Exit and reflection | Keep the roles distinct

Short sentences or labelled arrows are welcome.

E 1. One barrier and its protective action.

E 2. What kind of thing is an antibody?

E 3. Why can a response to the same antigen be faster later?

L1. What did your audience question? Record one useful revision or defended explanation.

Next useful step

- A. Explain a barrier action.
- B. Separate cells from proteins.
- C. Connect memory with specificity.
- D. Read matched times on a graph.